GradientCIO

PORTFOLIO ATTRIBUTION COMPARISON

# Northstar Attribution Comparison

Historical, forward and simulated views

PORTFOLIO
**Northstar Total Fund**

VIEWS
**Three**

AUDIENCE
**Investment Committee**

DATA AS OF
**2026-10-08**

OVERALL VIEW
**Three compatible lanes compared**

SCOPE REVIEWED
**4 of 4 · Complete**

# 01 Executive Comparison

Three compatible governed lanes have server-returned comparison metrics

OVERALL VIEW

**Three compatible lanes compared**

SCOPE REVIEWED · 4/4

BOTTOM LINE

Historical, ex ante, and simulated effects are compatible, and governed comparison metrics are available S2 S3 S4 S5.

| HISTORICAL LANE | EX ANTE LANE | SIMULATED LANE | COMPARISON COVERAGE |
|---|---|---|---|
| **Available** | **Available** | **Available** | **Complete** |
| Realized 12 months · 2025-07-01 to 2026-06-30 S2 | 1-year expected S3 | 1-year simulated · governed scenario S4 | Server-returned comparison · compatible receipts S5 |

## Three governed returned lanes and comparison metrics are available S2 S3 S4 S5.

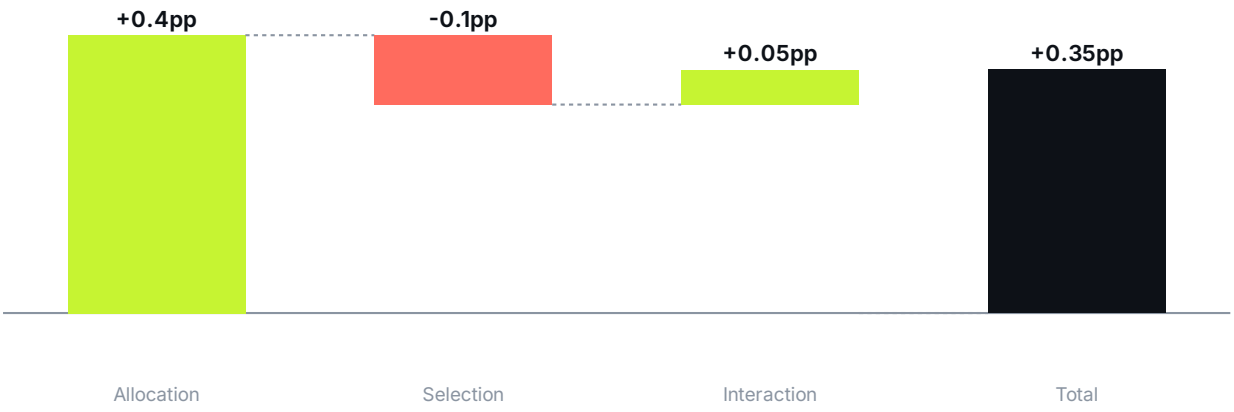
Returned evidence only; no local subtraction or accumulation.

REALIZED SUMMARY EFFECTS




Each plotted value is a returned summary field S2.

EXPECTED SUMMARY EFFECTS




Each plotted value is a returned summary field S3.

**Key judgment — Historical**

Observation: Historical effects are returned S2. Why it matters: Realized sources can be described. Uncertainty: One period may not persist. What would change the view: A later governed period.

**Key judgment — Ex ante**

Observation: Ex ante effects are returned S3. Why it matters: Expected sources can be described. Uncertainty: Inputs are assumptions. What would change the view: Refreshed governed inputs.

**Key judgment — Comparison**

Observation: Server-returned comparison metrics cover three compatible lanes S5. Why it matters: Cross-lane evidence is governed. Uncertainty: One comparison receipt limits time-series interpretation. What would change the view: A later receipt returning different lane signs.

## 02 Scope and Compatibility

Portfolio, benchmark, cohort and currency identifiers match

### Compatible for governed three-lane comparison

The returned compatibility receipt matches portfolio, benchmark role, parent cohort and calculation currency S5.

| | | |
|---|---|---|
| **Compatibility gate** | **available** | Four identifiers match S2 S3 |

## 03 Historical Attribution

Returned realized effects are shown without recalculation

**Historical effects by segment**

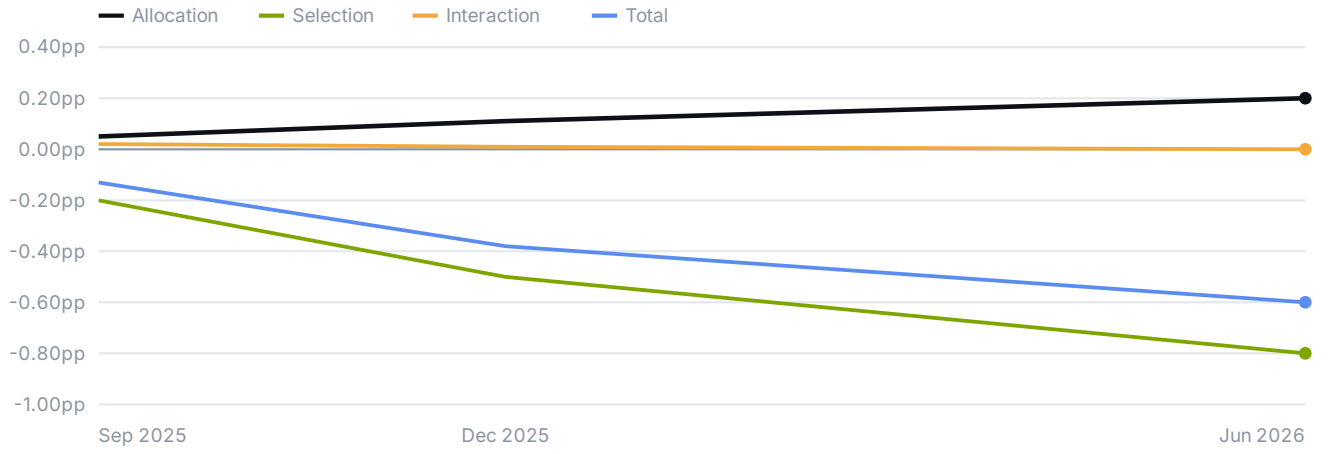
Not available — stacked values contain no positive total

HISTORICAL PORTFOLIO AND BENCHMARK AVERAGE WEIGHTS



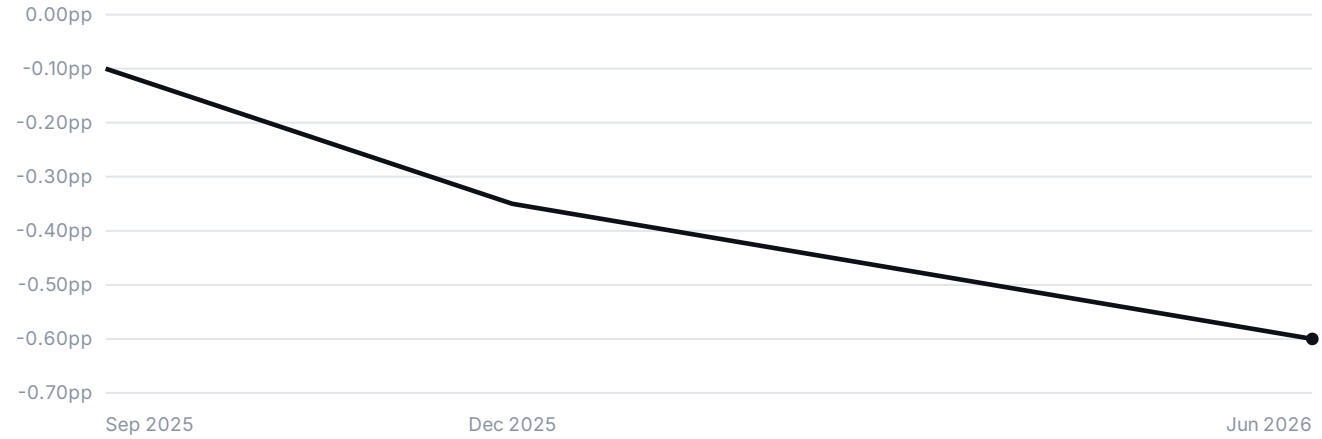

Returned arithmetic average beginning weights; no local difference S2.

CUMULATIVE LINKED EFFECTS



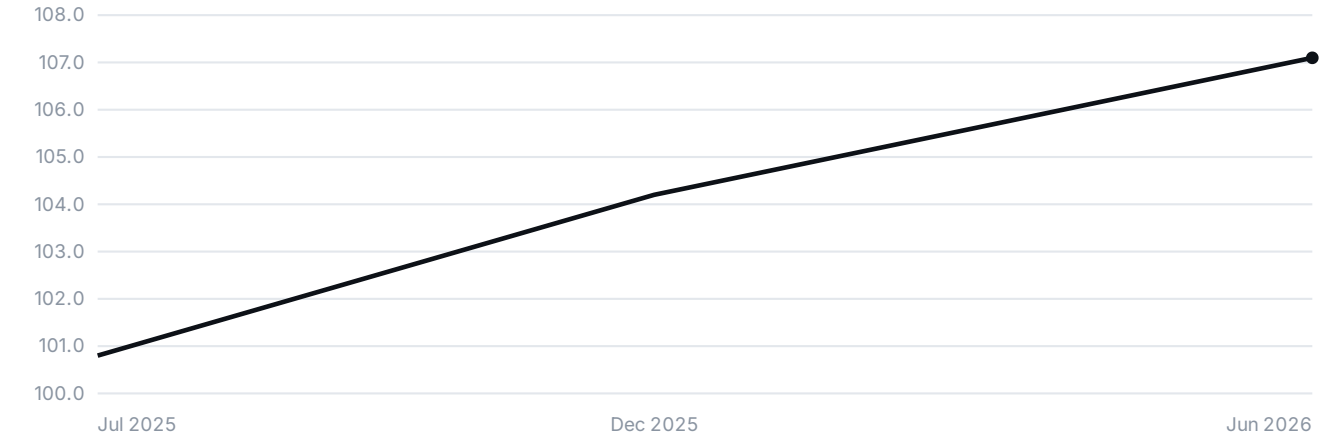

Server-returned cumulative linked fields; no local accumulation S2.

CUMULATIVE ALTERNATIVES SELECTION




Returned rows filtered to allocation_name=Alternatives; no local accumulation S2.

RETURNED CUMULATIVE GROWTH CONTEXT




Optional context rendered only from returned cumulative_growth rows S6.

**Analysis — Selection leads detractors**

Observation: Returned selection is negative S2. Why it matters: Selection leads realized detractors. Uncertainty: The evidence covers one linked period. What would change the view: A later historical result.

**Benchmark and excess boundary**

Returned diagnostics.entity_return_reconciliation keeps attribution-summary excess against the linked segment benchmark composite separate from historical-return excess against the policy benchmark entity S2.

## 04 Governed Ex Ante Attribution

Returned expected effects are shown as assumptions

EXPECTED EFFECTS BY SEGMENT

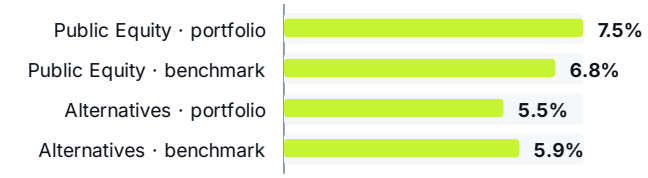



Returned expected allocation, selection and interaction by segment S3.

## EX ANTE ASSUMPTIONS

| | |
|---|---|
| assumption_set_name | **2026 Strategic CMA** S3 |
| regime_name | **user-defined regime** S3 |
| horizon | **1 year** S3 |
| portfolio_return_source | **allocation_expected_return_base** S3 |
| benchmark_return_source | **benchmark_expected_return_base** S3 |

## EXPECTED PORTFOLIO AND BENCHMARK RETURNS




Returned expected returns grouped without local subtraction S3.

**Analysis — Allocation leads expectations**

Observation: Returned allocation is positive S3. Why it matters: Allocation leads expected effects. Uncertainty: Inputs are assumptions. What would change the view: A refreshed ex ante result.

# 05 Simulated Attribution

Governed simulation returns scenario effects

### SIMULATED EFFECT BANDS




Server-returned scenario bands and selected values S4.

### SCENARIO SELECTOR

| SCENARIO ID | SCENARIO | SELECTED | BASIS |
|---|---|---|---|
| scenario-base-test | Base scenario | Yes | 1-year simulated |
| scenario-down-test | Downside scenario | No | 1-year simulated |
| scenario-up-test | Upside scenario | No | 1-year simulated |

Returned scenario selector; no locally invented scenarios S4.

**Capability gate** — available — Exact simulation capability was available before invocation S1 S4

**Analysis — Scenario range returned**

Observation: The governed simulation returned effect bands and a selected base scenario S4. Why it matters: Scenario context is explicit. Uncertainty: Results depend on governed scenario inputs. What would change the view: A revised simulation receipt.

## 06 Cross-Lane Comparison

Governed comparison returns compatible cross-lane metrics

### RETURNED CROSS-LANE EFFECT MATRIX

| | REALIZED | EXPECTED | SIMULATED |
|---|---|---|---|
| ALLOCATION | 0.20 | 0.40 | 0.30 |
| SELECTION | -0.80 | -0.10 | -0.20 |
| INTERACTION | 0.00 | 0.05 | 0.04 |

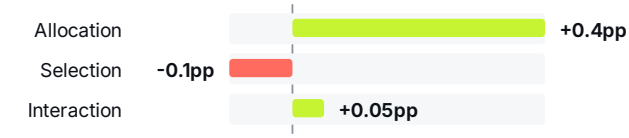
Server-returned three-lane effect matrix S5.

### REALIZED RETURNED EFFECTS



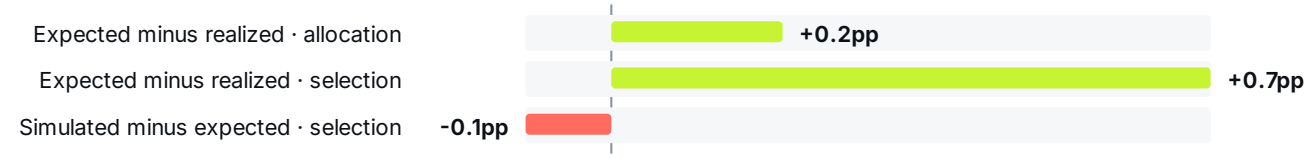

Displayed unchanged S2.

### EXPECTED RETURNED EFFECTS



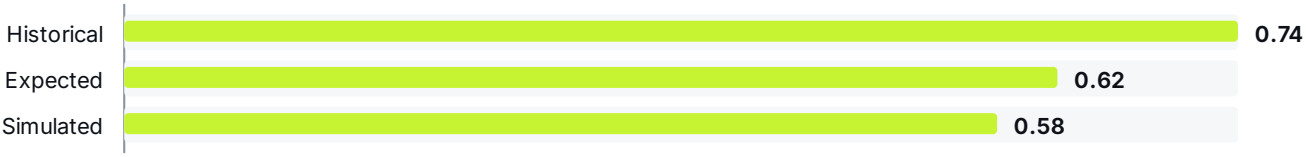

Displayed unchanged S3.

### RETURNED LANE DIFFERENCES




Differences are returned by the comparison tool, not calculated here S5.

### RETURNED CONCENTRATION




Server-returned concentration scores S5.

### RETURNED PERSISTENCE

| EFFECT | RETURNED LANES | PERSISTENCE STATUS |
|---|---|---|
| Allocation | Historical · Expected · Simulated | Persistent |
| Selection | Historical · Expected · Simulated | Persistent negative |
| Interaction | Expected · Simulated | Partial |

Server-returned persistence labels S5.

> **Analysis — Governed comparison fields returned**
>
> Observation: The server returned differences, concentration and persistence for three compatible lanes S5. Why it matters: Cross-lane claims use governed fields. Uncertainty: One receipt limits time-series interpretation. What would change the view: A later comparison receipt returning different statuses.

## 07 Coverage and Limitations

Two live lanes and two capability gaps define coverage

| | | |
|---|---|---|
| **Historical** | available | Governed result S2 |
| **Ex ante** | available | Governed result S3 |
| **Planned tools** | available | Capability-gated results S4 S5 |

# A Appendix — Sources and methods

Governed source inventory

| TAG | EVIDENCE | BASIS | TOOL / VIEW | KEY PARAMETERS | AS OF | DATA SCOPE | VALIDATION | DIGEST |
|---|---|---|---|---|---|---|---|---|
| S1 | Capability gate | gradient_tool | get_gradient_capabilities | illustrative sample | 2026-06-30 | Gradient-governed review evidence | passed | sha256:aaaaaaaaaaaa |
| S2 | Historical attribution | gradient_tool | get_portfolio_attribution | illustrative sample | 2026-06-30 | Gradient-governed review evidence | passed | sha256:bbbbbbbbbbbb |
| S3 | Ex ante attribution | gradient_tool | get_portfolio_ex_ante_attribution | illustrative sample | 2026-06-30 | Gradient-governed review evidence | passed | sha256:cccccccccccc |
| S4 | Simulated attribution | gradient_tool | get_gradient_capabilities | illustrative sample | 2026-06-30 | Gradient-governed review evidence | passed | sha256:dddddddddddd |
| S5 | Cross-lane comparison | gradient_tool | get_gradient_capabilities | illustrative sample | 2026-06-30 | Gradient-governed review evidence | passed | sha256:eeeeeeeeeeee |
| S6 | Historical return context | gradient_tool | get_portfolio_historical_returns | fields=cumulative_growth; limit=100; illustrative sample | 2026-06-30 | Gradient-governed review evidence | passed | sha256:ffffffffffff |

Illustrative, Gradient Maintained — demo data, not the client's holdings or managers. All organizations, portfolios, managers and results in this public example are illustrative.

Full digests — S1: sha256:aaaaaaaaaaa